A Steganographic Algorithm for Hiding Data in PDF Files Based on Equivalent Transformation

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Abstract

To amend the defects of low perceptual transparency, low capacity, low security of the present information hiding technology in PDF files, we present a novel steganographic algorithm which is better in perceptual transparency property, capacity and security. Attributive to the insensitiveness of PDF files to the seriation of entries, the algorithm helps to realize the information hiding in high degree of perceptual transparency and in large capacity without writing new data in the original file. And as the initial value of the chaotic map, the key is generated by applying the secure hash algorithm, to keep the system meet Kerckhoffs rules and enable integrity service. In addition, this algorithm is equipped strong anti-statistical attack ability.

1. Introduction

1.1. Background and related work

People have long been studying diverse ways to hide information from anyone except for the information receiver when sending out messages. Thus, there was the birth of cryptology. In encryption schemes, Alice turns the information into cryptograph by encryption and send it to the receiver Bob. Without the key, the third party is unable to read the message even if the one has already intercepted it. However, the encryption information alone is practically insufficient and easily triggers the suspicion from the third party. Therefore, a way to hide the communication itself is required, which leads to the development of information hiding technology. While cryptology is trying to hide the information, the steganography is advancing to cover the communication of information which is underway, thus enabling the two corresponding parties to exchange information secretly. In steganographic model, Alice and Bob represent the two parties exchanging secret information in a covert way. In hope of sending the secret message to Bob, Alice randomly chooses a cover-object as a vehicle, embeds secret information in it with a stego-key, and then sends it to Bob. In this way, the third party won't be able to detect the secrecy they are delivering even if he discovers the vehicle cover-object [2,3].

In this algorithm, PDF text is adopted as the cover-object. Its application independent from hardware and operation systems, PDF file can be transmitted and shared conveniently in different computer platforms [1]. As a result, it has developed into the mainstream file format in network communication. However, the relevant study in this bearing is rare. In the present PDF-based algorithms of information hiding, steganographic methods are proposed almost by varying the line or word or character spacing or by varying certain character features slightly [4,6,8,10,11,12]. Nevertheless, the defects that the effect of page display is disturbed, that the original size of the cover-object is changed and that information security is relatively low are present in the above methods. In addition, by embedding about 2 bytes of information between two objects, the software wbStego4.3 [9] helps to hide information in PDF files. However, its capacity is small and hiding data in a PDF file will increase its file size.

1.2. Introduction of the algorithm suggested

In this paper, the method based on equivalent transform is adopted in this algorithm to hide information. In this way, some merits can be developed, for example, that page display and file size will not be changed, and that the capacity is enlarged. Meanwhile, by treating the key generated from SHA-1 as the initial value of logistic chaotic map, it ensures data integrity and the randomicity of equivalent transform and keeps the whole mechanism in line with Kerckhoffs rules.

1.3. Paper outline

The rest of this paper is organized as follows. Section2 introduces the covert channel in PDF files. Section3 presents the algorithm. The experiment results and analysis are given in Section 4. Followed by conclusions and future work in Section 5.

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2. Covert channel in PDF files

PDF file structure can be classified into physical structure and logical structure: the physical structure includes the header, the body which contains a lot of objects, the cross-reference table containing information about the indirect objects in the file and the trail. The logical structure of PDF file is organized in the shape of an object tree topped by Catalog, Page tree, Outline hierarchy and Article thread included. The Outline hierarchy is the bookmarker of PDF, and Page tree includes page and Pages which in turn includes the total page number and each page marker. Page, the main body of PDF file, is the most important object which involves the typeface applied, the text, pictures, page size and so on. The organization of other objects is analogous to Page tree. With the object tree topped by Catalog, any object in PDF file can be visited.

An object is the basic element in PDF files, in which eight kinds of objects, namely Boolean Object, Numeric Object, String Object, Name Object, Array Object, Null Object, Dictionary and Stream Object are sustained. Our main concern goes to Dictionary and Stream Object which both contain dictionary. Dictionary is formed by coupling of a series of property pairs named dictionary entries, the first element of which is key is specified to be a name, and the second element of which is value can be any object, or even another dictionary. After abundant empirical study, it is discovered that the effect of page display of PDF file is extraneous to the seriation of entries. Thereby, covert information embedded in different objects can be achieved by special array of entries, instead of by operation of adding any other data to the cover-object. Moreover, the large number of existing dictionaries and stream objects in PDF files ensures the large capacity required of covert information embedded.

3. Principle and algorithm

3.1. Principle of information hiding

Definition 1: The object that can embed information in the PDF file is called valid object.

Definition 2: Let $\alpha_n = (a_1, a_2, \cdots, a_n)$ be a sequence of n entries in one valid object in PDF file. Where the form of a_i is key-value pair, $1 \le i \le n$ and $n \ge 2$. Arrange $(a_1, a_2, \cdots, a_n)$ in the sequence in the order of 0 to $n!$ according to the lexicographic order [5], $\alpha_n^k = (a_1, a_2, \cdots, a_n)$ is a sequence with a serial number of k, while $k \in \mathrm{Z}$ and $0 \le k \le n!-1$. As the alternation of the sequence of elements within the vectors does not lead to the change of the size of PDF file and the display of the page, the α_n^k is called the equivalent transform of α_n, and the symbol is $\alpha_n^k \Leftrightarrow \alpha_n^k$.

Definition 3: Let $\mathbf{O}_j(\alpha_{n_j})$ be the jth valid object in PDF file, the sequence of its entries is α_{n_j}, $\mathbf{O}_j(\alpha_{n_j}^k)$ is the new object after the equivalent transform of α_{n_j} into α_n^k, as $\alpha_{n_j}^k$ is the equivalent transform of α_{n_j}, $\mathbf{O}_j(\alpha_{n_j}^k)$ to $\mathbf{O}_j(\alpha_{n_j}^k)$ does not change the size of PDF file and the display of the page, $\mathbf{O}_j(\alpha_{n_j}^k)$ is called a equivalent transform object of $\mathbf{O}_j(\alpha_{n_j}^k)$ and is labeled $\mathbf{O}_j(\alpha_{n_j}^k) \Leftrightarrow \mathbf{O}_j(\alpha_{n_j})$.

Definition 4: Let $E_j = \left\{\mathbf{O}_j(\alpha_{n_j}^k) \middle| k \in \mathrm{Z}, 0 \le k \le n!-1\right\}$ be the set of all equivalent transform objects of $\mathbf{O}_j(\alpha_{n_j})$, set $\mathrm{Z}_j = \left\{z \middle| z \in \mathrm{Z}, 0 \le z \le n_j - 1\right\}$, the mapping $|\bullet|$ from E_j to Z_j is defined as $\left|\mathrm{O}_j(\alpha_{n_j}^k)\right| = k$; it is know from Definition 2, the mapping is bijective, and the inverse mapping is $|k|^{-1} = \mathrm{O}_j(\alpha_{n_j}^k)$.

Definition 5: If $\left|\mathrm{O}_j(\alpha_{n_j}^k)\right| = t_j$, the number of valid objects is M, let $T = (t_1 \cdots t_j \cdots t_M)$, $\mathbf{n} = (n_1 \cdots n_i \cdots n_M)$; Defining function $f(T, \mathbf{n}) = \sum_{j=2}^{M}\left[\prod_{i=1}^{j-1}(n_i!)t_j\right] + t_1$. Then function $f(T, \mathbf{n})$ is bijection, which mapped a group of equivalent transform objects to the embedded information.

Theorem 1: If I is the maximum capacity of the PDF file with M valid objects, then $I = \log_2^{\left(\prod_{j=1}^{M} n_j!-1\right)}$.

Prove: That is, to obtain the value of $f(T, \mathbf{n})_{\max}$, obviously when $T = (n_1!-1, n_2!-1, \cdots, n_M!-1)$, $f(T, \mathbf{n})$ has the maximum value:

$$f(T, \mathbf{n})_{\max} = \sum_{j=2}^{M}\left[\prod_{i=1}^{j-1}(n_i!)(n_j!-1)\right] + n_1!-1$$

$$= \prod_{j=1}^{M} n_j!-1$$

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Then $I = \log_2^{\left(\prod_{j=1}^{M} n_j! - 1\right)}$

3.2. Logistic chaotic map

Logistic Chaotic Map [7] is defined as:

$x_{n+1} = \mu x_n (1 - x_n), \mu \in [0,4], x \in [0,1]$;

μ is called fractal parameter, when $\mu \in (0.357, 4)$, the systems enters into chaos, the average value of the produced actual worth sequence is 0.5, the ergodicity is equaled to the white noise with an average of 0.5.

Take $x_n = \mathrm{sgn}(x_k) = \begin{cases} 1, & x_k > 0.5 \\ 0, & x_k < 0.5 \end{cases}$

It produces binary sequence $\{x_k | \ k = 1,2,3\cdots, x_k \in \{0,1\}\}$, the algorithm of this paper uses this random binary sequence produced by logistic chaotic map to equivalently transform entries by obeying the following rules:

Embedding:

Step 1:Let $k = 1$

Step 2: Let $s = x_k \oplus x_{k+1}$, if $s = 1$, then $\alpha_n^{'} = (a_1, \cdots a_{k+1}, a_k, \cdots a_n) \Leftrightarrow \alpha_n$; If $s = 0$, then no changes should be made.

Step 3:Let $k = k + 1$; If $k = n$, then end the random equivalent transform of α_n, otherwise goto Step 2;

Extracting:

Step 1: Let $k = n$

Step 2: Let $s = x_k \oplus x_{k-1}$; If $s = 1$, then $\alpha_n^{'} = (a_1, \cdots a_k, a_{k-1}, \cdots a_n) \Leftrightarrow \alpha_n$; If $s = 0$ then no changes should be made;

Step 3: Let $k = k - 1$; If $k = 1$, then end the extraction, otherwise goto Step 2.

3.3. Key generation

According to the characteristics of fact that Logistic chaotic map is sensitive to the initial value, the initial x_0 is taken as the key. The algorithm of this paper uses SHA-1 to produce 160 bits digest with the input of embedded information, four bits of the 160-bits digest fall within a group, and 40 decimal integer is thus produced, which when are connected according the order, a decimal fraction with 0 at the place of the integer. It is taken as the initial value of x_0, the 160 bits digest is the key of the system. As the information is extracted, the comparison of the hashed value of the extracted information to the key could verify the integrity of data.

3.4. Embedding algorithm

Figure1 shows the overview of the data embedding steps, and the following is the text description of the embedding algorithm.

Step 1: Input the embedded data to produce the key and the initial value x_0;

Step 2: Ransack the PDF file, to obtain E_j with every $\mathbf{O}_j(\alpha_{n_j}), 1 \le j \le M$;

Step 3: Obtain the maximum capacity $N_{\max} = \prod (n_j!) - 1$;

Step 4: Change the binary bit string which is to be embedded to integer N, if $N_{\max} < N$, then the embedding fails; otherwise Let $j = 1$, goto Step 5;

Step 5:Let $I^{'} = I/(n_j!)$, $t_i = I \bmod (n_j!)$;

Step 6: Let $I = I^{'}$, $j = j + 1$, if $j \le M$, goto Step 5, otherwise goto Step 7;

Step 7: Generate T, make equivalent transform to every valid object in cover-object according to T.

Step 8: To produce Logistic chaotic map sequence with x_0 as the initial value, make equivalent transform to every valid object again according to the embedding rule as section3.2 describes to scramble the sequence of entries.

Step 9: Output the PDF file and key.

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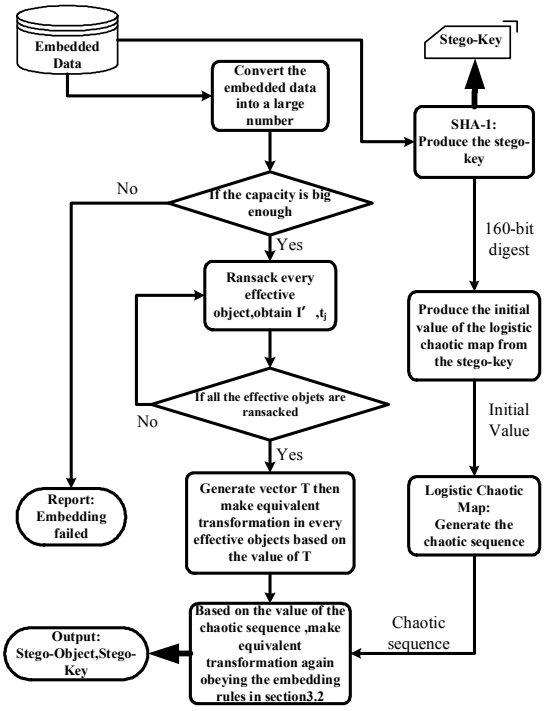

Figure1. An overview of data embedding steps

3.5. Extracting algorithm

Figure2 shows the overview of the data extracting steps, and the following is the text description of the extracting algorithm.

Step1: Input the key, and then generate the chaotic sequence with the initial value x_0;

Step2: Ransack the PDF file, obtain M and $\mathbf{n}$, carry on the restoration of each valid object in accordance with the extraction rule as section3.2 describes, and Let j=1;

Step3: according to $\mathbf{O}_j(\alpha_{n_j}^k)$, obtain $\left|\mathrm{O}_j(\alpha_{n_j}^k)\right| = t_j$;

Step4: Let j=j+1,if $j \le M$, goto Step3, otherwise goto Step5;

Step5: obtain $T' = (t_1, t_2, \cdots, t_M)$, because mapping $|\bullet|$ is bijective mapping, $T' = T$, meanwhile, the value of the n has nothing to do with the Algorithm, and then $f(T', \mathbf{n}) = N' = f(T, \mathbf{n}) = N$;

Step6: produce 160-bits digest through SHA-1, if it is the same with the key, then output extraction information, otherwise the data is changed or destructed in the dissemination process.

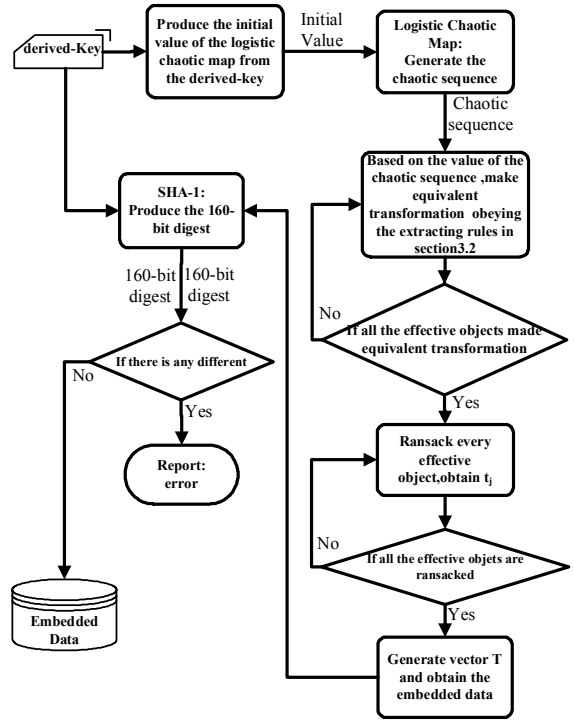

Figure 2. An overview of data extracting steps

3.6 Security analysis

In this paper, the key is regarded as the initial value of the logistic chaotic map, and it is used in producing the random sequence to carry on the scrambling. Because of the logistic chaotic map's sensitivity to initial value, the security of embedding and extraction algorithm obeys the Kerckhoffs principle. At the same time, because the embedding algorithm does not change the display effect of the cover-object and the document size, the algorithm has the very strong anti-statistical attack ability.

4. The result and analysis of the experiment

4.1 Perceptual transparency property

In our experiment, the cover-object we have chosen is Adobe SDK ReleaseNotes.pdf. Lena picture with size of 2309bytes is treated as the coming embedding data. The comparison of effects before the data embedded and those after the data embedded shown in the Figure 3.

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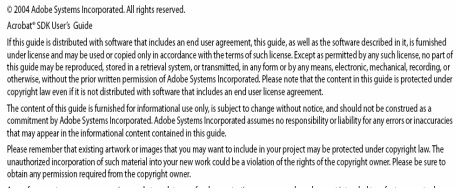
Table1. The result of the performance comparison

Performance	Equivalent Transformation	wbStego4.3	The method based on varying characters spacing
Perceptual Transparency	No changed	No changed	Slightly changed
Embedding Capacity	Large	Small	Large
Increase in the size of cover-object	No	No	Yes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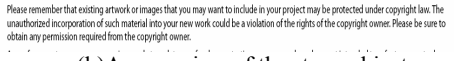
(a)An overview of the cover-object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(b)An overview of the stego-object

Figure3. The cover-object and the stego-object

Seen from the effects chart, after having embedded data, there was not any change in display effect of the cover-object. The algorithm didn't write any new data in the cover-object, and wouldn't change the size of the document.

4.2 The performance comparison

Table1 shows the performance comparison of different steganographic methods .Though the algorithm based on the varying the line or word or character spacing, has realized the large capacity of data, they shift the display effect of cover-object, simultaneously also changed the size of cover-object. In this case, it was unable to prevent the general statistical attack effectively, and the security is low. But in this paper, our algorithm not only doesn't change the display effects of the cover-object, but also doesn't make any revision to the document size. It has the characteristics of anti-statistical attack.

5. Conclusion and future work

This paper proposed a novel PDF document steganographic algorithm, in which equivalent objects transformation is used to embed data to the cover-object. The analysis and experimental results indicated that the algorithm obtain perceptual transparency and high security. How to guarantee the safety to prevent various attacks is the emphasis for our next step work.

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